Tate's theorem for a class formation
The splitting-module argument

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Private reading transcription of the author's Theorem II.3.11, with the complete supplied proof of its supporting Theorem II.3.10 and the separately published correction to that proof. The original work is retained at https://www.jmilne.org/math/CourseNotes/CFT.pdf. The selected material occupies physical PDF pages 92–94 (printed pages 83–85). The correction is retained with the official errata page in the source package. Mathematical expressions and the author's argument are retained; connective prose and layout are normalized. This is published-material transcription, not author-native TeX or a new proof.

Notation and the author's convention

The group G is finite. In this part of Section II.3 the author uses Tate cohomology throughout, dropping the subscript T except in the main statements. We preserve that convention in the transcribed proofs. In positive degrees these groups agree with ordinary group cohomology. In degree zero the convention is

$$H^0_T(H,M) = M^H/\operatorname{Nm}_H M, \qquad \operatorname{Nm}_H(m) = \sum_{h\in H} hm.$$

The action on $\mathbb{Z}$ is trivial. The notation M_0 in the supporting argument denotes the underlying abelian group with trivial action, as in the source's preceding induced-module conventions (II.1.3–1.4). The preceding cohomological tools used by the author are cyclic periodicity, the exact cohomology sequence, the double boundary map (II.1.13), restriction/corestriction (II.1.33), inflation/restriction (II.1.34), and the Tate-acyclicity of induced modules. These are context for the supplied construction, rather than additional counted problems.

The target statement

Theorem II.3.11 (Tate's theorem). *Let G be a finite group and C a G-module. Suppose that for every subgroup $H \le G$, including $H = G$,*

1. *$H^1(H,C) = 0$;*
2. *$H^2(H,C)$ is cyclic of order $(H:1)$.*

Then for every integer r there is an isomorphism

$$H^r_T(G,\mathbb{Z}) \longrightarrow H^{r+2}_T(G,C)$$

depending only on the choice of a generator for $H^2(G,C)$.

The supporting all-degree vanishing theorem

Theorem II.3.10 (support). *Let G be a finite group and M a G-module. If*

$$H^1(H, M) = 0 = H^2(H, M)$$

for all subgroups H of G, then $H^r_T(G, M) = 0$ for all $r \in \mathbb{Z}$.

Author proof as printed on p.83. For cyclic G the conclusion follows from periodicity. For solvable G, argue by induction on its order. There is a proper normal subgroup H for which G/H is cyclic. The pair (H, M) satisfies the hypotheses, so induction gives $H^r(H, M) = 0$ for all r. By II.1.34 there are exact sequences, for $r \geq 1$,

$$0 \longrightarrow H^r(G/H, M^H) \longrightarrow H^r(G, M) \longrightarrow H^r(H, M). \qquad (*)$$

Because $H^1(G, M) = H^2(G, M) = 0$, these give $H^1(G/H, M^H) = H^2(G/H, M^H) = 0$. Cyclic periodicity for G/H then gives $H^r(G/H, M^H) = 0$ for all r. The sequences $(*)$ imply $H^r(G, M) = 0$ for $r > 0$.

To show $H^0(G, M) = 0$, let $x \in M^G$. Since $H^0(G/H, M^H) = 0$, there is $y \in M^H$ such that $\mathrm{Nm}_{G/H}(y) = x$. Since $H^0(H, M) = 0$, there is $z \in M$ with $\mathrm{Nm}_H(z) = y$. Hence

$$\mathrm{Nm}_G(z) = (\mathrm{Nm}_{G/H} \circ \mathrm{Nm}_H)(z) = x.$$

Thus $H^r(G, M) = 0$ for every $r \geq 0$.

To proceed further, tensor the augmentation sequence (19) with M, obtaining

$$0 \longrightarrow M' \longrightarrow \mathbb{Z}[G] \otimes_{\mathbb{Z}} M \longrightarrow M \longrightarrow 0.$$

By II.1.4 and II.1.3, respectively,

$$\mathbb{Z}[G] \otimes_{\mathbb{Z}} M \simeq \mathbb{Z}[G] \otimes_{\mathbb{Z}} M_0 \simeq \mathrm{Ind}^G(M_0)$$

is induced. Therefore $H^r(H, \mathbb{Z}[G] \otimes_{\mathbb{Z}} M) = 0$ for every r and every subgroup H of G, and the cohomology sequence gives

$$H^r(H, M) \simeq H^{r+1}(H, M')$$

for all r and all H. In particular, M' satisfies the hypotheses of the theorem. The part already proved gives $H^r(G, M') = 0$ for $r \geq 0$, so

$$0 = H^0(G, M') = H^{-1}(G, M).$$

Repeating this argument gives $H^{-2}(G, M) = 0$, and so on. This proves the theorem for solvable G.

For an arbitrary finite G, its Sylow p-subgroup G_p and M also satisfy the hypotheses. The printed proof concludes that $H^r(G_p, M) = 0$ for all r and p, and, using II.1.33, that the p-primary component of $H^r(G, M)$ is zero for all r and p. It then concludes $H^r(G, M) = 0$ for all r. □

Separately published correction to II.3.10

The official errata page records Kiran Kedlaya's correction to the preceding proof. The use of H^r in place of H^r_T obscures the degree-zero issue in the reduction to solvable groups. That reduction as invoked in the printed Sylow argument is valid for $r > 0$, while degree zero is also needed for shifting to negative degrees.

The correction supplies the missing degree-zero step. Since $H^0_T(G_p, M) = 0$, the group $H^0_T(G, M)$ is killed by $[G : G_p]$. Varying p gives $H^0_T(G, M) = 0$; dimension shifting then gives the negative-degree conclusions. This paragraph records the correction supplied on the author's errata page; it is separate from the printed proof above.

The author's proof of II.3.11

Author proof. Choose a generator γ of $H^2(G, C)$. Because $\mathrm{Cor} \circ \mathrm{Res} = (G : H)$, the element $\mathrm{Res}(\gamma)$ generates $H^2(H, C)$ for every subgroup H of G.

Let φ be a cocycle representing γ. Define $C(\varphi)$ as the direct sum of C and the free abelian group with basis x_σ, one for each $\sigma \in G$, $\sigma \neq 1$. Extend the G-action on C by setting

$$\sigma x_\tau = x_{\sigma\tau} - x_\sigma + \varphi(\sigma, \tau).$$

Here x_1 is interpreted as $\varphi(1, 1)$, exactly as in the source. This defines an action, since

$$(\rho\sigma)x_\tau = x_{\rho\sigma\tau} - x_{\rho\sigma} + \varphi(\rho\sigma, \tau),$$

whereas

$$\begin{aligned}\rho(\sigma x_\tau) &= \rho\big(x_{\sigma\tau} - x_\sigma + \varphi(\sigma, \tau)\big) \\ &= x_{\rho\sigma\tau} - x_\rho + \varphi(\rho, \sigma\tau) - \big(x_{\rho\sigma} - x_\rho + \varphi(\rho, \sigma)\big) + \rho\varphi(\sigma, \tau).\end{aligned}$$

These expressions agree by the cocycle condition

$$\rho\varphi(\sigma, \tau) + \varphi(\rho, \sigma\tau) = \varphi(\rho\sigma, \tau) + \varphi(\rho, \sigma).$$

The cocycle φ is the coboundary of the 1-cochain $\sigma \mapsto x_\sigma$, so γ maps to zero in $H^2(G, C(\varphi))$. For this reason $C(\varphi)$ is called the splitting module for γ.

We first show that the hypotheses imply

$$H^1(H, C(\varphi)) = 0 = H^2(H, C(\varphi))$$

for every subgroup H of G. The augmentation sequence is

$$0 \longrightarrow I_G \longrightarrow \mathbb{Z}[G] \longrightarrow \mathbb{Z} \longrightarrow 0,$$

where I_G is free abelian on $\sigma - 1$, with $\sigma \in G$, $\sigma \neq 1$. The module $\mathbb{Z}[G]$ is induced, so $H^r(H, \mathbb{Z}[G]) = 0$ for all r. Consequently

$$\begin{aligned}H^1(H, I_G) &\simeq H^0(H, \mathbb{Z}) \simeq \mathbb{Z}/(H : 1)\mathbb{Z}, \\ H^2(H, I_G) &\simeq H^1(H, \mathbb{Z}) = 0.\end{aligned}$$

Define the additive map $\alpha : C(\varphi) \longrightarrow \mathbb{Z}[G]$ by

$$\alpha(c) = 0 \quad (c \in C), \qquad \alpha(x_\sigma) = \sigma - 1.$$

The resulting sequence of G-modules

$$0 \longrightarrow C \longrightarrow C(\varphi) \xrightarrow{\alpha} I_G \longrightarrow 0$$

is exact. Its cohomology sequence reads

$$\begin{aligned}0 \longrightarrow H^1(H, C(\varphi)) \longrightarrow H^1(H, I_G) \longrightarrow H^2(H, C) \\ \xrightarrow{0} H^2(H, C(\varphi)) \longrightarrow 0.\end{aligned}$$

The end zeros use $H^1(H, C) = 0$ and $H^2(H, I_G) = 0$. The indicated map from $H^2(H, C)$ is zero because that group is generated by $\mathrm{Res}(\gamma)$, whose image is the restriction of the image of γ in

$H^2(G, C(\varphi))$, already zero. Thus $H^1(H, I_G) \longrightarrow H^2(H, C)$ is onto, and is an isomorphism because both groups have the same finite order. Its kernel and cokernel, $H^1(H, C(\varphi))$ and $H^2(H, C(\varphi))$, both vanish.

By Theorem II.3.10, $H^r(H, C(\varphi)) = 0$ for all r. Splicing the two short exact sequences gives

$$0 \longrightarrow C \longrightarrow C(\varphi) \longrightarrow \mathbb{Z}[G] \longrightarrow \mathbb{Z} \longrightarrow 0. \tag{24}$$

Both middle terms are Tate-acyclic: for every r, $H^r(G, C(\varphi)) = 0 = H^r(G, \mathbb{Z}[G])$. The double boundary map is therefore an isomorphism, by II.1.13,

$$H^r(G, \mathbb{Z}) \longrightarrow H^{r+2}(G, C).$$

□

The single collected target is II.3.11. The supporting theorem, its published correction, and the auxiliary exact sequences add no separate problem count. The generator is chosen, rather than asserted canonical. The source's following Remark II.3.13 identifies the map as cup product with that chosen element; this reader retains the supplied splitting-module proof and does not add an independent proof of the remark or the later local-field application.